

Section 8.4: Stage 3 - Development and Testing

When designs for the new system (stage 2) have been complete, development can begin. Development is stage 3.

A system's developer will follow the designs to produce a working system that meets all requirements.

There are two parts to stage 3.... **Development** and **Testing**.

Key Concepts of this section:

- # Understand what is **involved** in the **development and testing** stage.
- # Know the **different stages** that are involved during development and testing.
- # Be able to describe the **different stages** that are involved during development and testing.

Stage 3a - Development

What is system development?

- # System development just means to **'build the system'**.
- # When stage 1 (analysis) and stage 2 (design) have been completed, there will be **clear lists of requirements and sets of designs** that a **developer can follow** in order to create a working system.
- # If analysis and design have been done properly, the developer should be able to create a system that works fully and is free of errors.

HINT - Think of systems development in the same terms as **building a house**.

A bricklayer follows instructions (plans) provided by an architect and a system's developer follows instructions provided by a system's analyst and designer.

- * The **architect** will include things like **wall length, height and shape** on a building's plans
- * The **systems analyst/designer** will include things like **calculations, layout of the input/output screens** and **how data will be entered** in the system plans.



Key Words:

Development, File Structure, Validation Checks, Data Entry Forms, Output Formats

Examples:



A builder will build a house correctly by reading plans provided by an architect.

A system is built in much the same way -- the developer will follow plans provided by systems' analysts and designers.

What are the stages of development?

The development stage is broken down into **4 parts**:

1. **Creating a file structure** to store data
2. **Create validation rules** to make sure that data entered is sensible
3. **Create a user-interface** to allow data to be entered into the system
4. **Create output formats** (reports, payslip's, bills etc)

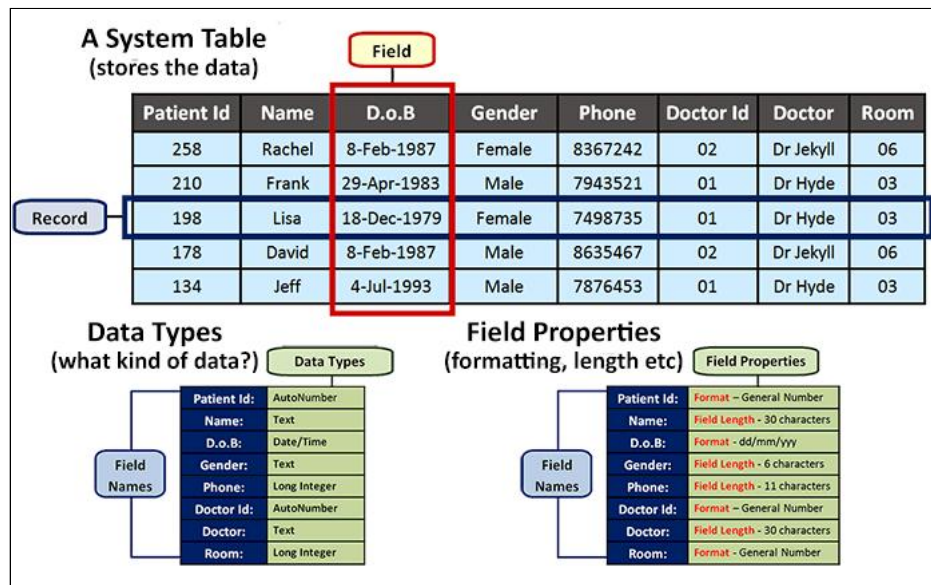
1 Creating file structures to store data

Systems (like databases) that are required to **hold data** need to use **file structures**.

File structures determine **how the data is stored** in the system and are made up of 4 parts:

- | | | | |
|---------------|--------------------------------------|------------------------------------|----------------|
| 1. | 2. | 3. | 4. |
| Tables | Fields | Field Properties | Records |
| | (each field is assigned a data type) | (field length, decimal places etc) | |

An example of a file structure can be seen in the image below:
(For more details see [Unit 5 - Database Structures](#))



Test data is entered into the fields to make sure that **data is being stored correctly**.

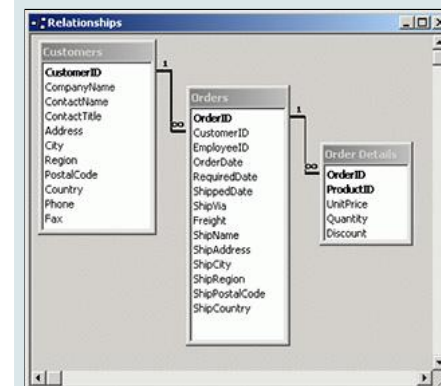
Examples:

Most systems are required to hold lots of data. File structures allow this data to be stored in an organised way (click to zoom)

NOTE:

If the system makes use of **more than one table**, the developer will need to also need to set up **key fields** in order to **link** all of the tables together.

This was covered in detail during [Unit 5 - Types of Databases](#)



A system with more than one table is linked together using key fields (the bold text)

2 Creating Validation Rules / Checks

Validation checks will have been thought about and decided on during the design stage.

REMEMBER:

Validation checks make sure that only **correct data** can be entered into the system.

For example:

A validation check of **Male OR Female** set as part of the **Gender field** in a database system would ensure that the user could not input anything else.

If the user tried to enter **anything other** than '**Male**' / '**Female**', the system would flag this up as an **error**.

Once all of the necessary validation checks have been programmed into the system, they are **tested** to make sure that they work.

Validation checks are tested in two ways:

1. Entering data that **should be accepted**
2. Purposefully entering data that **should not be accepted**.

IF the validation check is working, the **acceptable data** will be **entered** into the system **without problem** and the **unacceptable data will be rejected**.

Examples:

Field Name	Data Type
First Name	Text
Gender	Text

Field Name	Field Properties
Gender	Field Size: 1, Format: , Input Mask: , Caption: , Default Value: , Validation Rule: ""F" Or "M"", Validation Text: Please enter either "F" or "M", Required: No, Allow Zero Length: No

A validation rule which will force users to only enter 'F' (for female) or 'M' (for male) .

First name	Gender
	G

Microsoft Office Access

Please enter either "F" or "M"

OK Help

The validation rule in action. I have tried to enter 'G' into the gender field and the system has rejected the data

3 Creating Input Methods (Data Entry Forms)

User-interfaces have to be created to allow users to **input data into the system**.

The screens of a system that allow users to enter data into the system are called '**Data entry forms**'. Data entry forms should be well designed to make sure that they are easy to use.

A good data entry form should....

- * Include **entries** for all of the **required fields**
- * Make use of **form controls** appropriately (see below)
- * Include **instructions** which tell the user how to enter data correctly
- * Make sure that **text boxes** are **large enough** to enter the required data
- * Be **well designed** and easy to understand

Examples:

What are form controls?

- # Form controls are the **items** on a data entry form that **allow you to interact with the system** and **enter data**. Form controls include:

Enter your name in the box below: Entering here.....	Text boxes Used to enter text using a keyboard
Select your gender from the options below: ☒ Male ☐ Female	Option buttons Used to select from a number of options (only one option can be selected)
Select your employment benefits from the options below: ☒ Insurance ☒ Healthcare ☐ Company Car	Check boxes Used to select from a number of options (more than one option can be selected)
Select your employment benefits from the options below: Healthcare Insurance Healthcare Company Car	Combo boxes (drop-down lists) Used to select one item from a list of options
Save Close Open	Command buttons Used to perform a command (save for example)

Tour Request Form

Complete the form below and someone will contact you with tour details, costs and scheduling options!

Contact Name

Title

Organization

Email

Phone
 - -
Type of Organization
☒ College Organization
☐ Other Student Organization
☐ Community Group
☐ Conference or Event
We'd ideally like you to come sometime between this date...
 / /
...and this date
 / /
Please briefly describe your organization and event.

Submit

A data entry form.

NOTE:

The data entry form makes use of **form controls appropriately**.

For example - questions where the **answer could be anything** (like name) are assigned **text boxes**.

Questions with **limited choices** (like type of organisation) are **restricted to 4 options**

4 Creating Output Formats (reports, invoices, payslip's etc)

- # Two types of system outputs have to be created:

- On-screen outputs
- Printed outputs (hard copies)

Examples:

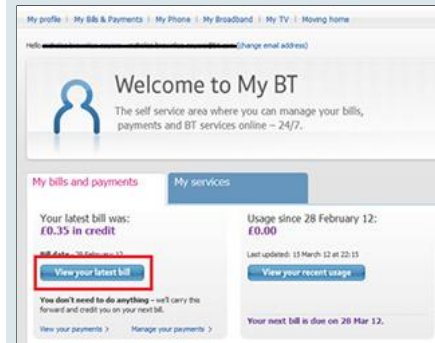
- # The system developer will create all outputs by following designs provided during stage 2.
- # The developer will ensure that any **hardcopy outputs** will be **printable** by clicking a '**Print**' button.
- # Outputs are **tested** by checking that the information output is **clear, easy to read, complete** and **correct**.

The image below shows a system report being designed:

The screenshot shows a report design window titled 'rptEmployees : Report'. It features a grid-based layout with sections for Report Header, Page Header, Detail, Page Footer, and Report Footer. The Report Header section contains the title 'Employees Report'. The Page Header section contains fields for First Name, Last Name, Gender, and Birth Date. The Detail section contains fields for First Name, Last Name, Gender, and Birth Date. The Page Footer section contains a date field set to '=Now()'. The Report Footer section contains two summary rows: 'Male Employees:' with the formula '=Count(IIf([Gender]="M",0))' and 'Female Employees:' with the formula '=Count(IIf([Gender]="F",0))'.



Some outputs are designed to be printed
(like this phone bill).



Some outputs are designed to be displayed on-screen
(like this online phone bill)..

NOTE:

On-screen outputs can be created with **features** that cannot be included with printed outputs.

For example - on-screen outputs can make use of **animation, hyperlinks, sound, video** (multimedia) etc.

Printed outputs are limited to good old **text, images, colours** and **charts** etc.

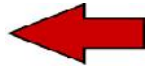
Stage 3b - Testing

Coming soon.....



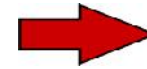
Key Words:

Examples:



[Previous - Design](#)

Next - Integration



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 2: Input and Output Devices](#)

[Section 3: Storage Devices and Media](#)

[Section 4: Computer Networks](#)

[Section 5: Data Types](#)

[Section 6: The Effects of Using ICT](#)

[Section 7: The Ways in which ICT is used](#)

[Section 7: Part 2](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 10: Document Production](#)

[Section 11: Data Manipulation](#)

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